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Safety and immunogenicity of a 25-valent pneumococcal conjugate vaccine in pneumococcal vaccine-naive healthy adults: Results from 2 randomised, controlled clinical trials

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A R T I C L E I N F O

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A B S T R A C T

Background: Pneumococcus causes substantial morbidity and mortality worldwide in children under 5. IVT PCV-25 is a 25-valent pneumococcal conjugate vaccine (PCV25) designed to prevent invasive pneumococcal disease from the serotypes predominant in children, particularly in low and middle income countries (LMICs).
Methods: We completed 2 randomised, parallel-group, double-blind clinical trials in Canada to evaluate the safety and immunogenicity of a single IM dose of PCV25 in healthy adults who had no history of pneumococcal vaccination or microbiologically confirmed IPD. PCV20 (Prevnar 20) was the control. In CVIA 096, 30 participants per group were randomised to PCV25 at a dose similar to PCV 20 (2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 125 μg aluminium as aluminium phosphate (2.2/125)) or PCV20. Potentially more immunogenic formulations with higher polysaccharide and/or aluminium phosphate dose were evaluated in CVIA105, where 40 participants were randomised to PCV25 (2.2/125), 60 to PCV25 (2.2/250), 80 to PCV25 (4.4/250), and 40 to PCV20.
Results: Most participants were female and white. All participants were included in the safety analyses. One participant in CVIA 096 and 6 participants in CVIA 105 were excluded from the immunogenicity analyses because of protocol deviations that might interfere with immune response. Solicited and unsolicited AE profiles were similar for PCV25 and PCV20. No Grade 4 events were reported. At the highest dose, PCV25 elicited IgG and OPA responses with GMFRs of ≥2 for all 25 serotypes and for serotype 6A except for OPA response to 35B.
Conclusions: Multiple formulations of IVT PCV-25, a vaccine designed to cover pneumococcal serotypes prevalent in LMICs, were well tolerated and immunogenic in healthy adults. As adult immunogenicity is not fully predictive for infants, further development will evaluate safety and immunogenicity in the target infant population.
ClinicalTrials.gov NCT05540028, NCT06077656

Abbreviations: CCfV, Canadian Center for Vaccinology; HZ-PEG-HZ, hydrazide-polyethylene glycol-hydrazide; LMICs, low- and middle-income countries; NDCMC, newly diagnosed chronic medical condition.

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1. Introduction

The first pneumococcal conjugate vaccine (PCV), PCV7, containing 7 capsular polysaccharides individually conjugated to the carrier protein CRM_{197} (a nontoxic mutant of diphtheria toxin) was introduced for use in infants and young children in 2000 in the US [1] and subsequently replaced by vaccines with higher valency. As of May 2025, 170 of 194 WHO member states have introduced PCVs for infant vaccination, and 15 more are planning to do so [2] because of their significant reduction of morbidity and mortality in children and the indirect protection observed in the wider population [3].

A meta-analysis evaluating the global impact of PCV10 and PCV13 use estimated 45 to 85% reductions in pneumococcal pneumonia in children up to 9 years and reductions in all-cause and severe pneumonia hospitalisations [4]. Effectiveness against vaccine-type invasive pneumococcal disease (IPD) has been reported for PCV10 and PCV13 for both the 3 + 1 (86 to 96% and 73 to 100%, respectively) and 2 + 1 schedules (67 to 86% and 92 to 97%, respectively) [5].

However, despite the introduction of effective PCVs, pneumococcus remains a leading cause of otitis media, sepsis, meningitis, and pneumonia in children worldwide [6]. Up to 95% of the estimated 225,000 deaths in children under 5 years of age attributable to pneumococcal infection occur in Asia and Africa [7]. IPD cases continue to occur, and the rate of non-vaccine-type IPD has increased dramatically [8] [9]. Higher valency PCVs have been licensed and more are under development but these primarily target high-income markets.

The cost of PCVs, compared to that of other routinely administered childhood vaccines, has always been a concern for public health programs. Multivalent PCVs are expensive to produce, and LMICs that do not qualify for external financial support (such as from Gavi, The Vaccine Alliance) can rarely afford these vaccines within their national health budgets. A less expensive PCV10 (Pneumosil) achieved WHO prequalification in 2019, but its effectiveness is limited by the increasing burden of disease caused by non-vaccine serotypes. The Gates Foundation is funding Inventprise Inc. and PATH to develop effective vaccines that are affordable for global use.

The candidate PCV25, Inventprise's IVT PCV-25, is intended to prevent invasive and noninvasive pneumococcal disease of the remaining predominant serotypes that cause disease in children, particularly those residing in African countries and in other LMICs (Fig. 1, Table 1). Most of the capsular polysaccharides from the serotypes in PCV13 and PCV20 are included in IVT PCV-25 because of their established global importance as invasive serotypes, and the additional serotypes in IVT PCV-25 (2, 6C, 9 N, 15 A, 16F, 24F, and 35B) were selected because of their identification as the most common invasive serotypes globally. Serotype 6 A which is included in PCV 13 and PCV20 was not included because of the expected cross-protection from Serotypes 6B and 6C, a serotype unique to PCV25 that is a more common cause of invasive pneumococcal disease than 6 A [20]. Serotype 11 A, which is included in PCV20 was not included in IVT PCV-25 because of its low invasive potential in children [21].

Analysis of global IPD serotype distribution suggests that IVT PCV-25 would address more than 80% of paediatric IPD in all regions [20]. IVT PCV-25 uses CRM_{197} as the protein conjugate and a novel hydrazide-polyethylene glycol-hydrazide (HZ-PEG-HZ) linker that increases immune response to the pneumococcal polysaccharides in animals, perhaps because of an increase in half-life of the polysaccharide conjugate, and reduced steric hindrance by the protein carrier due to the linker length and polysaccharide sizing [22]. Both binding and functional serotype-specific antibody responses to the 18 common serotypes were higher for PCV25 than for PCV20 in these pre-clinical studies, and responses to the additional 7 serotypes in PCV25 were similar to the responses to the 18 common serotypes [22]. Although vaccine performance in adults is not necessarily predictive for infants, the studies described in this paper were conducted to provide an initial assessment of safety and immunogenicity prior to evaluation in the target infant population.

Table 1
Rationales for pneumococcal serotypes included in IVT PCV-25.

Serotype(s)	Rationale for inclusion
1, 3, 4, 5, 6B, 7F, 9 V, 14, 18C, 19 A, 19F, 23F	Most likely to cause IPD globally, included in PCV13, PCV15, and PCV20
2	Highly invasive serotype associated with IPD in several LMICs, specifically Bangladesh, Guatemala, and Israel [10,11]
6C	6C is substituted for 6A because of the cross-protection between serotype 6A and 6B [12]. It is an increasing cause of invasive antimicrobial-resistant, pneumococcal disease.
8	Cause of IPD in children <5 years of age since 2010 in Gavi and non- Gavi countries, included in PCV20
9N	Invasive serotype in children <5 years of age, no cross-protection seen by serotype 9V seen in currently licensed PCVs [13]
10A	One of the most common invasive serotypes in in paediatric and adult populations in both high-income countries and LMICs, included in PCV20
12F	A highly invasive serotype known to cause epidemic disease, included in PCV20
15A, 15B	Common cause of IPD in children <5 years of age, especially in non- Gavi countries, 15A included in PCV20 and associated with significant antimicrobial resistance [14] [15]
16F	Increasing cause of IPD in <5 years of age in Gavi and non- Gavi countries [16]
22F, 33F	Invasive, included in PCV15 and PCV20
24F	Emerging invasive serotype described as the most common cause of meningitis in France as well as reports from Argentina, Germany, Papua New Guinea and Peru [17] [18]
35B	Emerging invasive serotype, associated with multidrug resistance [19]

IPD, invasive pneumococcal disease; LMICs, low- and middle-income countries; PCV, pneumococcal conjugate vaccine.

2. Methods

We conducted 2 randomised, double-blind, parallel-group, clinical trials to evaluate the safety, reactogenicity, and immunogenicity of a single dose of PCV25 with PCV20 as an active control in healthy, pneumococcal vaccine-naive adults.

Both studies were conducted in compliance with Good Clinical

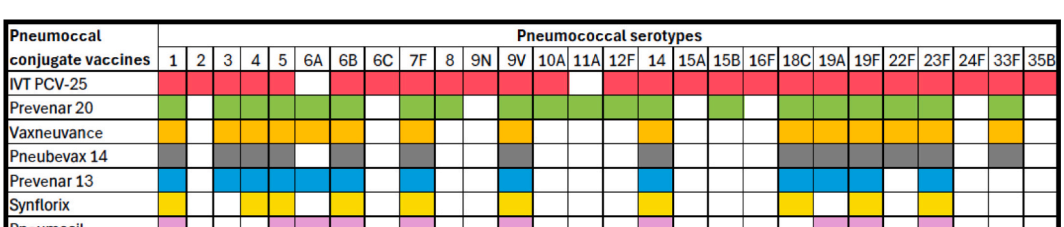

Fig. 1. Pneumococcal serotypes in IVT PCV25 and PCVs licensed in children.

Practice guidelines, the Declaration of Helsinki, Canadian regulatory requirements and were approved by local institutional review boards and the WIRB-Copernicus Group (CVIA 096: 21 Sep 2022, IRB tracking: 20224609; CVIA105: 07 Sep 2023; IRB tracking 20,233,981). The protocol and statistical analysis plan for each study can be accessed at ClinicalTrials.gov. Informed consent was obtained from each participant after the nature and consequences of the study had been fully explained to the participant. There was no patient or public involvement in the design, conduct, or reporting of the trial.

2.1. Overall study design and vaccines

The Phase 1 study (CVIA 096; ClinicalTrials.gov NCT05540028, registered 09 September 2022) was conducted from November 2022 to July 2023 by the Canadian Center for Vaccinology (CCfV) in Halifax, Canada. Healthy adults aged 18 to 40 years who had no history of pneumococcal vaccination or microbiologically confirmed IPD were randomised 1:1 to receive a single dose of PCV25 or PCV20 by intramuscular (IM) injection using a 22 to 25 gauge, 1.0- or 1.5-in. needle into the mid-deltoid muscle of the participant's non-dominant arm on Day 1. Both vaccines were administered at a dose of 2.2 μg for each serotype except 4.4 μg for serotype 6B with 125 μg aluminium as aluminium phosphate; PCV20 was administered in accordance with the approved product monograph. This study was initially designed to proceed to paediatric cohorts after enrolment of the adult cohort; however, after review of group-level, unblinded immunoglobulin G (IgG) results in a planned data snapshot, Inventprise and PATH decided to amend the trial to not enrol the paediatric cohorts but to conclude the adult cohort as planned and proceed to evaluation of other formulations in adults.

The Phase 2, dose-ranging study (CVIA 105; ClinicalTrials.gov NCT06077656, registered 10 October 2023) was conducted from October 2023 to June 2024 by 4 sites in Canada: the CCfV, the Colchester Research Group, the Vaccine Evaluation Center at British Columbia Children's Hospital Research Institute, and the Centre de Recherche – Saint Louis. This study evaluated IVT PCV 25 at a dose similar to PCV20, and two formulations with higher polysaccharide dose, as has been used in the recently licensed PCV-21 and in other PCVs in development. One of the higher polysaccharide dose formulations also contained a higher dose of aluminium phosphate adjuvant which may facilitate optimal antigen adsorption. Of note the higher dose is lower than that used in most licensed aluminium adjuvanted vaccines [23].

Healthy adults aged 18 to 49 years who had no history of pneumococcal vaccination or microbiologically confirmed IPD were randomised 2:3:4:2 to receive a single dose of PCV25 in 1 of the 3 following formulations or PCV20 at the approved dose by IM injection at the same site on Day 1:

- Formulation A (2.2/125): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 125 μg aluminium as aluminium phosphate adjuvant (same formulation as used in CVIA 096)
- Formulation B (2.2/250): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate adjuvant
- Formulation C (4.4/250): 4.4 μg for each serotype polysaccharide (except 8.8 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate adjuvant

For both studies, PCV25 (IVT PCV-25; Inventprise Inc.) was provided in multi-dose vials with 2-phenoxyethanol preservative (Formulation A: Lots 22E004, 230512002; Formulation B: Lot 230,512,003; Formulation C: Lot:230512001). PCV20 (Prevnar 20; Pfizer Canada ULC) was commercially available vaccine provided in prefilled syringes product sourced in Canada (Lots GF1709, GM1337). No other vaccines were to be administered within 14 days before study vaccine administration until after the Day 29 blood draw. Both vaccines were stored at 2 to 8 °C in dedicated refrigerators connected to a power source with a reliable back-up system that were under continuous temperature monitoring with maintenance of daily temperature logs. Randomisation was performed via an interactive web response system (IVRS) according to randomisation schemes generated by a statistician at the CCfV data and statistics group; unblinded study site staff who used the IVRS to assign participants to vaccine groups had access to the random allocation sequence. Fixed block randomisation ($N = 4$ in CVIA 096, $N = 11$ in CVIA 105) was employed to ensure that the randomisation ratio was maintained between the vaccine groups. Study staff were not aware of block contents or sizes. Unblinded study staff who were not involved in outcome assessment recorded participant randomisation and prepared and administered the vaccines. The participants, the study site personnel performing postvaccination study assessments, and laboratory personnel were blinded.

2.2. Primary and secondary endpoints

The primary objective in both studies was to assess the safety and tolerability of PCV25 administered as a single-dose regimen in healthy pneumococcal vaccine-naive adults through 28 days post-vaccination. The endpoints were solicited local and systemic adverse events (AEs) during the first 7 days post-vaccination, clinically significant haematological and biochemical measurements at 7 days post-vaccination (CVIA 096 only), unsolicited AEs through 28 days post-vaccination, serious AEs (SAEs) through 6 months post-vaccination; and newly diagnosed chronic medical conditions (NDCMCs) through 6 months post-vaccination (CVIA 096 only).

The secondary objectives in both studies were to evaluate serotype-specific IgG responses and functional antibody responses against the 25 *S pneumoniae* serotypes in PCV25 plus serotype 6 A and compare the antibody responses induced by each vaccine against these serotypes at 28 days post-vaccination. The endpoints for IgG response were serotype-specific IgG geometric mean concentrations (GMCs) 28 days post-vaccination and the associated GMC ratios between the PCV25 and PCV20 groups, geometric mean fold rises (GMFRs) in serotype-specific IgG GMCs, and GMFR ratios between the PCV25 and PCV20 groups. Percentage of participants with a 4-fold rise in serotype-specific IgG from baseline to 28 days post-vaccination was also evaluated in CVIA 105 only. The endpoints for functional antibody response were serotype-specific opsonophagocytic assay (OPA) geometric mean titres (GMTs) measured 28 days post-vaccination and the associated GMT ratios between the PCV25 and PCV20 groups, GMFRs, and GMFR ratios between the PCV25 and PCV20 groups.

2.3. Safety and tolerability assessments

A complete physical exam was performed as part of determination of participant eligibility; thereafter a targeted physical exam was performed only in the event of a new symptom, sign, or AE. Solicited local and systemic AEs were assessed immediately (30 min) post-dose. Participants were provided with a thermometer for recording oral temperature, a ruler for measuring injection site reactions, and a paper diary for recording these measurements and for grading other solicited AEs (reactogenicity events) for 7 days post-dose. Solicited AEs were graded in accordance with the US Food and Drug Administration's guidance [24]. Unsolicited AEs were recorded through 28 days post-dose. Thereafter to the end of the study (24 weeks post-dose), only NDCMCs (CVIA 096 only) and SAEs were recorded. The injection site was assessed by study staff on Day 8. Vital signs were measured before and after vaccination on Day 1 and at the Day 8 and Day 29 visits. Blood samples for haematology (white blood cell count, haemoglobin, platelet count) and clinical chemistry (serum creatinine, alanine transaminase, aspartate transaminase, total bilirubin [CVIA 096 only]) were collected during screening and at the Day 8 visit and tested at each site's laboratory. A Data and Safety Monitoring Board reviewed unblinded safety after all participants completed the Day 29 visit.

2.4. Immunogenicity assessments

A serum sample was collected from each participant for immunogenicity evaluation before vaccination and at 28 days after vaccination.

For CVIA 096, serotype-specific IgG concentrations were measured by the UCL Institute of Child Health Vaccine Evaluation Laboratory (London, UK), a WHO pneumococcal serology reference laboratory, using an enzyme-linked immunosorbent assay (ELISA) validated for 16 serotypes (1, 3, 4, 5, 6A, 6B, 7F, 9V, 12F, 14, 18C,19A, 19F, 22F, 23F, 33F) and qualified for the other 10 serotypes. [25] For CVIA 105, IgG was measured at the same laboratory by an electrochemiluminescence immunoassay (ECLIA) on the Meso Scale Discovery platform which was validated for all serotypes. The ECLIA is a multi-array assay which has been formally bridged (cross-validated) to the WHO reference ELISA [26–28]. ECLIA permits the simultaneous detection and quantitation of up to 10 antigens resulting a significant reduction in reagents, consumables and serum volume.

Functional antibody responses were evaluated by Sunfire Biotechnologies (Birmingham, AL, USA) using a four-fold multiplexed OPA validated for 22 serotypes (1, 3, 4, 5, 6A, 6B, 6C, 7F, 8, 9N, 9V, 10A, 12F, 14, 15A, 15B, 18C, 19A, 19F, 22F, 23F, 33F) and qualified for the other 4 serotypes [29].

2.5. Statistical methods

The sample sizes in both studies were not based on any formal statistical hypothesis test. Probability of observing at least one AE according to different true event rates was evaluated in the protocols. For CVIA 105, the sample sizes in each group were weighted toward the higher antigen and adjuvant dose combinations to ensure safety assessment in each group at the interim analysis was adequate to advance to Phase 2 trial in infants and would reflect the probability for similar dose ranges to be included in the infant study. All analyses were descriptive.

Safety analyses were conducted in the safety analysis population (all participants who were randomised and received 1 dose and for whom any safety data were available). Frequencies and percentages of participants with any solicited local or systemic AE, solicited local event, solicited systemic event, unsolicited AE, NDCMC (CVIA 096 only), or SAE were tabulated along with an exact 2-sided 95% confidence interval (CI) computed via the Clopper-Pearson method. Frequencies and percentages of participants experiencing each solicited local or system event were presented by treatment group and severity. Frequencies and percentages of unsolicited AEs, related unsolicited AEs, NDCMCs (for CVIA 096 only), SAEs, related SAEs, and unsolicited AEs leading to participant discontinuation were summarised according to the Medical Dictionary for Regulatory Activities (MedDRA, Version 25.0 for CVIA 096 and Version 26.1 for Study 26.1). The number and percentage of participants with out-of-range (above and below normal range as appropriate) laboratory results were tabulated by toxicity grading and group, and summaries of changes from baseline were presented.

Immunogenicity analyses were performed in the per-protocol population (all participants who were randomised, received 1 dose, were correctly vaccinated per randomisation, had no protocol deviations determined before database unblinding to potentially interfere with the immunogenicity assessment of study vaccine, and for whom any post-administration immunogenicity results were available). Missing values were analyzed as if they were missing randomly and not imputed. GMCs at baseline and 28 days post-dose and GMFRs at 28 days post-dose were summarised for serotype-specific IgG antibody responses with 2-sided 95% CI for each treatment group. The same summary was provided for GMTs and GMFRs for serotype-specific functional antibody responses. The ratio of the GMC/GMT between each PCV25 dose group and PCV20 and the corresponding 2-sided 95% CI were provided. The log-transformed antibody responses were used to construct a mean difference between each PCV25 dose group and PCV20 and its 2-sided 95% CI using the 2-sample *t*-test method. The mean difference and corresponding 95% CI were back-transformed to obtain the GMC/GMT ratio between each PCV25 dose group and PCV20 and the corresponding 95% CI. An adjusted GMC/GMT ratio between each PCV25 dose group and PCV20 was also provided along with its 2-sided 95% CI. Adjusted models were performed using analysis of covariance with log-transformed baseline levels, age, and sex as covariates. The 2-sample *t*-test method was used to obtain the 2-sided 95% CI of the ratio of GMFRs between the study groups. Missing immunogenicity data were not imputed and were analyzed as if they were missing randomly.

Treatment groups for safety analyses were assigned according to the actual treatment received and for immunogenicity analysis according to the randomisation assignment. No randomisation errors occurred, so the treatment groups were the same for safety and immunogenicity analyses.

3. Results

3.1. Participants

A total of 280 participants were randomised in the 2 studies: 60 in CVIA 096 and 220 in CVIA 105. Most participants were female (71.2% in CVIA 096, 70.0% in CVIA 105) and White (88.3% in CVIA 096, 92.3% in CVIA 105) (Table 2). Median age was 34.0 years in CVIA 096 and 37.0 years in CVIA 105). All participants completed the study except 1 participant in the PCV20 group in CVIA 096 who was lost to follow-up after the Day 29 visit (Fig. 2). All participants were included in the safety analyses in both studies. One (2%) participant in CVIA 096 and 6 (3%) participants in CVIA 105 were excluded from the per-protocol

Table 2
Demographics characteristics.

Characteristic	Study CVIA 096			Study CVIA 105				
	PCV25 Formulation A n = 30	PCV20 n = 30	Total $n = 60$	PCV25 Formulation A n = 40	PCV25 Formulation B n = 60	PCV25 Formulation C n = 80	PCV20 n = 40	Total $n = 220$
Sex, n (%)								
Female	25 (83.3)	18 (60)	43 (71.7)	27 (67.5)	41 (68.3)	57 (71.3)	29 (72.5)	154 (70.0)
Male	5 (16.7)	12 (40)	17 (28.3)	13 (32.5)	19 (31.7)	23 (28.8)	11 (27.5)	66 (30.0)
Race, n (%)								
White	28 (93.3)	25 (83.3)	53 (88.3)	38 (95.0)	53 (88.3)	73 (91.3)	39 (97.5)	203 (92.3)
Asian	0	3 (10.0)	3 (5.0)	0	2 (3.3)	3 (3.8)	0	5 (2.3)
Black or African American	0	0	0	1 (2.5)	2 (3.3)	1 (1.3)	0	4 (1.8)
American Indian or Alaska Native	0	0	0	1 (2.5)	1 (1.7)	0	1 (2.5)	3 (1.4)
Multiple	0	1 (3.3)	1 (1.7)	0	1 (1.7)	2 (2.5)	0	3 (1.4)
Other	2 (6.7)	1 (3.3)	3 (5.0)	0	1 (1.7)	1 (1.3)	0	2 (0.9)
Age, years								
Mean (SD)	33.6 (4.17)	33.9 (4.12)	33.8 (4.12)	38.4 (6.78)	36.2 (6.40)	37.0 (7.83)	35.5 (8.38)	36.8 (7.40)
Median (range)	33.5 (23–40)	34.0 (25–40)	34.0 (23–40)	39.0 (25–49)	37.0 (24–48)	38.5 (21–49)	35.0 (20–49)	37.0 (20–49)

SD, standard deviation.
Formulation A (2.2/125): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 125 μg aluminium as aluminium phosphate.
Formulation B (2.2/250): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate.
Formulation C (4.4/250): 4.4 μg for each serotype polysaccharide (except 8.8 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate.

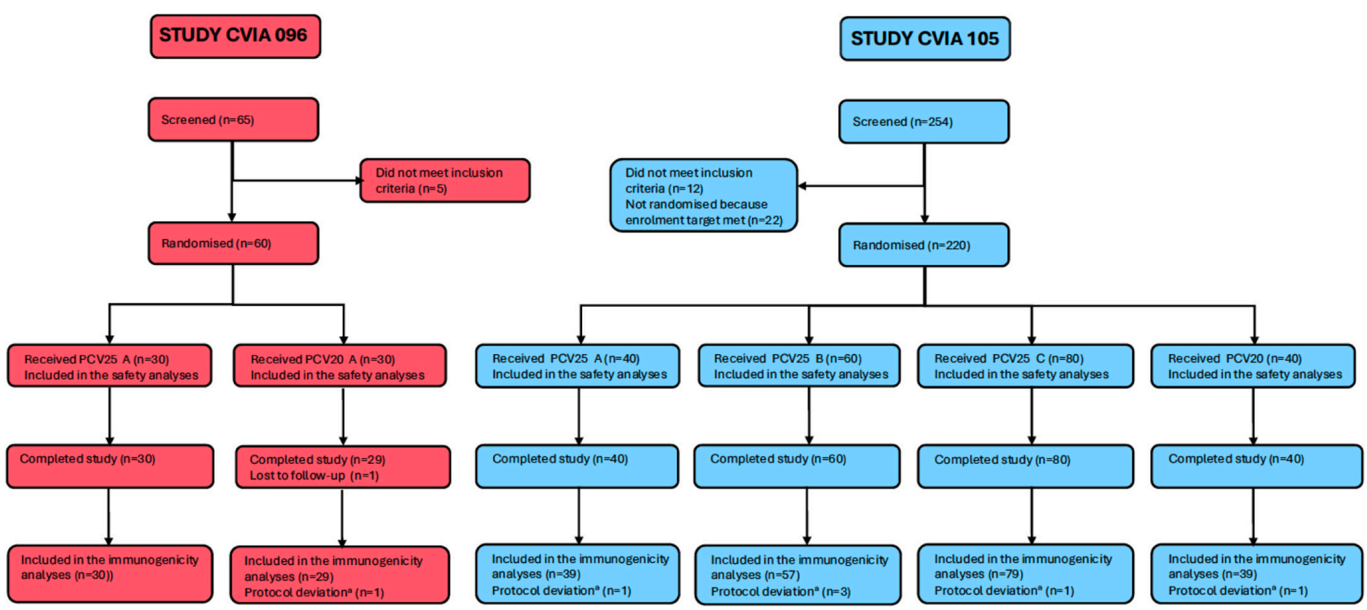

Fig. 2. CONSORT diagram. Formulation A (2.2/125): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 125 μg aluminium as aluminium phosphate Formulation B (2.2/250): 2.2 μg for each serotype polysaccharide (exce pt 4.4 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate Formulation C (4.4/250): 4.4 μg for each serotype polysaccharide (except 8.8 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate [a] Protocol deviation determined to potentially interfere with the immunogenicity assessment].

population used for the immunogenicity analyses because of protocol deviations determined to potentially interfere with the immunogenicity assessment.

3.2. Safety and tolerability

The safety and tolerability profiles for all outcome measures were similar for PCV25 and PCV20 (Table 3). No Grade 4 events were reported. One SAE was reported among the 280 participants in the 2 studies: severe appendicitis in a participant who received PCV25 Formulation C, assessed as not related to the study product, with onset on Day 144 and resolution by Day 163. No NDCMCs were reported.

Immediate solicited local and systemic AEs (within 30 min after injection) were reported by 9.0% of participants in the PCV25 groups and 5.7% of participants in the PCV20 groups in the 2 studies. The most frequently reported immediate event in the PCV25 groups was injection site pain and/or tenderness, which was reported by 4.3%, 10%, 6.3% of participants for the 3 PCV25 formulations and 5.0% for PCV20 group. Injection site redness, headache, arthralgia, and myalgia were also reported.

Table 3
Overview of solicited and unsolicited adverse events.

Category	Study CVIA 096		Study CVIA 105			
	PCV25 Formulation A n = 30 n (%) [95% CI]	PCV20 n = 30 n (%) [95% CI]	PCV25 Formulation A n = 40 n (%) [95% CI]	PCV25 Formulation B n = 60 n (%) [95% CI]	PCV25 Formulation C n = 80 n (%) [95% CI]	PCV20 n = 40[a] n (%) [95% CI]
Immediate solicited local or systemic adverse event	3 (10.0) [2.1, 26.5]	2 (6.7) [0.8, 22.1]	1 (2.5) [0.1, 13.2]	7 (11.7) [4.8, 22.6]	6 (7.5) [2.8, 15.6]	3 (7.5) [1.6, 20.4]
Local event	2 (6.7) [0.8, 22.1]	1 (3.3) [0.1, 17.2]	1 (2.5) [0.1, 13.2]	7 (11.7) [4.8, 22.6]	5 (6.3) [2.1, 14.0]	2 (5.0) [0.6, 16.9]
Systemic event	1 (3.3) [0.1, 17.2]	1 (3.3) [0.1, 17.2]	0	0	3 (3.8) [0.8, 10.6]	2 (5.0) [0.6, 16.9]
Solicited local or systemic adverse event	29 (96.7) [82.8, 99.9]	28 (93.3) [77.9, 99.2]	34 (85.0) [70.2, 94.3]	57 (95.0) [86.1, 99.0]	74 (92.5) [84.4, 97.2]	35 (89.7) [75.8, 97.1]
Local event	29 (96.7) [82.8, 99.9]	25 (83.3) [65.3, 94.4]	32 (80.0) [64.4, 90.9]	55 (91.7) [81.6, 97.2]	73 (91.3) [82.8, 96.4]	33 (84.6) [69.5, 94.1]
Grade 3	0	0	2 (5.0)	0	2 (2.5)	0
Systemic event	22 (73.3) [54.1, 87.7]	21 (70.0) [50.6, 85.3]	21 (52.5) [36.1, 68.5]	37 (61.7) [48.2, 73.9]	55 (68.6) [57.4, 78.7]	19 (48.7) [32.4, 65.2]
Grade 3	1 (3.3)	2 (6.7)	2 (5.0)	2 (3.3)	1 (1.3)	0
Unsolicited adverse event	12 (40.0) [22.7, 59.4]	8 (26.7) [12.3, 45.9]	17 (42.5) [27.0, 59.1]	24 (40.0) [27.6, 53.5]	30 (37.5) [26.9, 49.0]	16 (40.0) [24.9, 56.7]

CI, confidence interval.
Formulation A (2.2/125): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 125 μg aluminium as aluminium phosphate.
Formulation B (2.2/250): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate.
Formulation C (4.4/250): 4.4 μg for each serotype polysaccharide (except 8.8 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate.
[a] n = 39 for solicited adverse events.

Solicited local events were reported by most participants in the 7 days after vaccination: 90.0% in CVIA 096 and 87.7% in CVIA 105 (Table 3). The most common local event was injection site pain, which was reported by 90.0% of participants who received PCV25 and 82.9% of participants who received PCV20 (Supplementary Table 1). Grade 3 (severe) local events were reported by 4 (1.8%) of participants in the 2 studies, all in the IVT PCV-25 groups: 3 with injection site pain and/or tenderness and 1 with injection site swelling (Table 3, Fig. 3). The most common systemic events were myalgia (reported by 39.0% of participants who received PCV25 and 27.1% who received PCV20), headache (38.6% 35.7%), and fatigue (35.7%, 32.9%) (Fig. 3, Supplementary Table 1). Grade 3 (severe) systemic events were reported by 8 (3.6%) participants in the 2 studies: headache (1.9%, 0.5%), fatigue (1.0%, 2.9%), myalgia (0.5%, 0%), and decreased appetite (0.5%, 0%) (Fig. 3). Fever was rare. Mild fever (38.0 to 38.4 °C) was reported by 4 (1.9%) participants who received PCV25; for PCV20, 2 (2.9%) reported mild fever, and 2 (2.9%) reported moderate fever (38.5 to 38.9 °C) (Fig. 3).

No consistent pattern of toxicity or dose-dependent effect was seen for unsolicited AEs (Supplementary Tables 2 and 3). The most common events reported in the PCV25 groups were headache (reported by 4.3% of participants who received PCV25 and 7.1% who received PCV20) and upper respiratory tract infection (4.3%, 2,9%). Related events were reported for 3 (5.0%) of participants in CVIA 096: mild diarrhoea, moderate chills, and moderate dizziness, each in 1 participant in the PCV25 group. in CVIA 105, related unsolicited AEs were reported for 25 (13.9%) participants in the PCV25 groups and 3 (7.5%) in the PCV20 group. Most unsolicited AEs were mild or moderate in severity. AEs in 2.5% of participants were rated as severe, none of which were assessed as related to the study vaccine.

3.3. Immunoglobulin G responses

PCV25 was immunogenic for all 25 serotypes included in PCV25 and cross-reactive antibody was elicited for serotype 6A in both studies (Table 4). IgG GMC ratios were greater than 1.0 for all serotypes unique to PCV25, except for cross-reacting serotype 6C in the CVIA 105 Formulation A group. For the 18 common serotypes and cross-reacting serotype 6A which is not included in PCV25, IgG GMCs after vaccination in the PCV25 groups were generally lower than in the PCV20 group but were higher for serotypes 5, 6 A, 6B, 19F, and 23F in CVIA 096; for serotypes 5, 8, and 19F for all 3 PCV25 formulations in CVIA 105; and for serotypes 9V and 19A for the highest dose in CVIA 105. IgG GMFRs for the 18 common serotypes and for cross-reacting serotype 6A were also generally lower in the PCV25 groups than in the PCV20 group, but most GMFRs in the PCV25 groups were ≥ 4. GMFRs were generally higher with higher polysaccharide dose (CVIA 105 Formulation C) but there was no consistent trend with higher dose of aluminium phosphate adjuvant.

3.4. Functional antibody responses

OPA responses were generally similar to IgG (Table 5). For the 18 common serotypes, Day 29 OPA GMTs in the PCV25 groups were generally lower than in the PCV20 group but were higher for serotypes 5, 6B, and 19F in CVIA 096; for serotypes 5, 8, and 19F for all 3 PCV25 doses in CVIA 105; and for serotype 19A for the highest PCV25 dose in CVIA 105. OPA GMFRs for the 18 common serotypes and for cross-reacting serotype 6A were also generally lower in the PCV25 groups than in the PCV20 group, but most GMFRs in the PCV25 groups were ≥ 4. The majority of the unique serotypes elicited strong functional

Fig. 3. Solicited adverse events within 7 days postdose Formulation A (2.2/125): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 125 μg aluminium as aluminium phosphate Formulation B (2.2/250): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate Formulation C (4.4/250): 4.4 μg for each serotype polysaccharide (except 8.8 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate.

Fig. 3. (*continued*).

antibody responses but GMFR for serotype 35B was <2.0 for all formulations. Higher polysaccharide dose (CVIA 105 Formulation C) generally elicited stronger functional antibody responses but there was no consistent trend with higher dose of aluminium phosphate adjuvant.

4. Discussion

These first-in-human and dose-ranging studies assessed the safety and immunogenicity of a new PCV25 specifically designed to include serotypes relevant to the paediatric disease burden in LMICs and demonstrated an acceptable safety profile at all dose levels evaluated. All formulations were immunogenic, but the magnitude of immune response was somewhat lower than for the PCV20 comparator for most common serotypes. As was previously observed with other PCVs, such as with the increase from PCV13 to PCV20 [30], higher valency may be associated with reduced immunogenicity of some serotypes. Protein conjugation revolutionised vaccines' ability to elicit strong T-cell independent responses to polysaccharides, but increasing numbers of conjugated antigens or a history of prior immunisation against the carrier protein may interfere with immunogenic response likely due to CRM_{197}-mediated suppression and/or sharing limited T-cell help across multiple polysaccharide epitopes [31].

The WHO has established a population-based immune correlate of protection against IPD in infants from the pooled reverse cumulative distribution curve from 3 PCV7 efficacy studies [28]. Serotype-specific responder rates based on this threshold of IgG concentration ≥ 0.35 μg/mL is the primary endpoint for infant licensure, but no correlate has been established in adults. Other vaccines that induced lower GMCs, such as Synflorix, are associated with clinical effectiveness and reduction or elimination of vaccine-type disease [20]. The clinical immunogenicity and effectiveness of this candidate PCV25 vaccine needs to be determined in the target population, ie, infants.

Routine childhood PCV immunisation is well established in high-income countries, but PCVs are among the most expensive of childhood vaccines. Their inclusion in national infant immunisation programmes has been made possible in many LMICs by financial support from Gavi. Introduction of high-cost PCVs into Gavi-eligible countries may not be sustainable because countries graduating from Gavi support often cannot afford these vaccines and drop them from their publicly funded programmes [32]. IVT PCV-25 manufacturing process and cost of goods are optimized to make a broad coverage vaccine affordable for LMICs.

Some aspects of the study may limit its external generalisability. Although evaluation of safety is required first in adults before exposure of infants, the immunogenicity results in this population may not predict response in the target infant population. Adults are immunologically primed through many prior natural exposures to pneumococcus throughout life. Infants are naïve and have many immunologic differences to adults. Infant T-cells are skewed toward TH2 responses, have a higher proportion of regulatory (suppressor) subsets, and may be less likely to develop strong memory responses [33]. To overcome this tolerogenic immune milieu, infants receive 3 to 4 pneumococcal vaccinations to complete the series, a requirement that may produce a saturated immune response with a bell-shaped dose response curve when higher

Table 4

Immunoglobulin G geometric mean concentrations and geometric mean fold rises at 28 days post-dose a. Study CVIA 096.

a. Study CVIA 096

Serotype	GMC (95% CI), µg/mL		GMC Ratio (PCV25/PCV20) (95% CI)	GMFR (95% CI)	
	PCV25 Formulation A n=30	PCV20 n=29		PCV25 Formulation A n=30	PCV20 n=29
1	2.98 (1.90, 4.67)	4.10 (2.93, 5.75)	0.73 (0.42, 1.25)	7.29 (4.96, 10.73)	10.04 (6.93, 14.55)
3	0.34 (0.23, 0.51)	0.55 (0.34, 0.90)	0.62 (0.33, 1.14)	1.78 (1.32, 2.40)	2.34 (1.60, 3.44)
4	1.15 (0.77, 1.72)	1.63 (1.10, 2.41)	0.71 (0.41, 1.22)	4.80 (3.45, 6.67)	7.15 (4.68, 10.95)
5	3.51 (2.16, 5.71)	2.57 (1.57, 4.22)	1.37 (0.69, 2.69)	6.33 (3.54, 11.33)	8.59 (5.07, 14.54)
6B	10.66 (6.99, 16.24)	4.25 (2.35, 7.70)	2.51 (1.23, 5.10)	13.62 (9.28, 19.99)	8.91 (5.37, 14.77)
7F	2.91 (1.97, 4.32)	3.79 (2.51, 5.72)	0.77 (0.44, 1.34)	5.03 (3.48, 7.27)	7.51 (4.40, 12.81)
8	4.48 (3.15, 6.36)	7.14 (5.32, 9.57)	0.63 (0.40, 0.98)	3.21 (2.32, 4.44)	6.30 (4.34, 9.13)
9V	1.76 (1.11, 2.79)	2.79 (1.94, 4.01)	0.63 (0.35, 1.12)	4.93 (3.36, 7.22)	7.37 (4.38, 12.41)
10A	4.60 (2.84, 7.47)	10.34 (6.47, 16.51)	0.45 (0.23, 0.86)	3.95 (2.80, 5.56)	14.00 (9.12, 21.50)
12F	1.26 (0.92, 1.72)	1.43 (0.92, 2.24)	0.88 (0.52, 1.49)	4.14 (2.48, 6.91)	4.22 (2.88, 6.19)
14	4.56 (2.31, 8.98)	8.07 (4.75, 13.73)	0.56 (0.24, 1.32)	3.88 (2.51, 6.00)	6.95 (4.46, 10.82)
15B	10.45 (6.52, 16.74)	16.49 (11.27, 24.13)	0.63 (0.35, 1.15)	8.01 (5.53, 11.61)	14.17 (7.88, 25.48)
18C	3.51 (2.18, 5.64)	5.32 (3.47, 8.16)	0.66 (0.35, 1.24)	4.78 (3.28, 6.95)	12.07 (7.33, 19.87)
19A	5.65 (3.83, 8.33)	9.26 (5.88, 14.57)	0.61 (0.34,1.09)	2.71 (2.03, 3.62)	5.74 (3.47, 9.51)
19F	8.17 (5.23, 12.76)	5.95 (3.43, 10.33)	1.37 (0.69, 2.73)	6.46 (4.40, 9.49)	6.46 (3.85, 10.82)
22F	4.44 (3.12, 6.32)	8.61 (5.82, 12.73)	0.52 (0.31, 0.86)	4.49 (3.09, 6.50)	10.99 (6.21, 19.44)
23F	4.10 (2.63, 6.38)	2.85 (1.38, 5.86)	1.44 (0.63, 3.28)	9.97 (7.02, 14.17)	10.47 (5.96, 18.39)
33F	2.83 (1.78, 4.48)	7.02 (4.39, 11.22)	0.40 (0.21, 0.77)	4.29 (2.90, 6.35)	10.88 (6.89, 17.21)
2	14.12 (8.79, 22.69)	3.35 (2.11, 5.32)	4.22 (2.21, 8.06)	8.63 (5.44, 13.68)	1.65 (1.11, 2.47)
6C	20.05 (13.48, 29.80)	11.96 (7.35, 19.46)	1.68 (0.91, 3.08)	4.53 (2.93, 6.99)	3.10 (2.07, 4.63)
9N	2.53 (1.64, 3.90)	0.57 (0.39, 0.84)	4.43 (2.50, 7.85)	6.52 (4.73, 9.00)	1.88 (1.43, 2.48)
15A	14.79 (10.63, 20.59)	7.08 (4.99, 10.06)	2.09 (1.30, 3.35)	4.85 (3.22, 7.32)	2.77 (1.98, 3.87)
16F	10.10 (7.28, 14.02)	7.86 (6.12, 10.10)	1.28 (0.85, 1.95)	1.58 (1.25, 2.01)	1.27 (1.10, 1.47)
24F	12.40 (8.44, 18.22)	6.04 (4.47, 8.18)	2.05 (1.28, 3.29)	2.14 (1.69, 2.72)	1.13 (1.02, 1.26)
35B	9.59 (6.52, 14.09)	4.74 (3.12, 7.19)	2.02 (1.16, 3.52)	2.44 (1.85, 3.21)	1.04 (0.95, 1.13)
6A	5.01 (3.30, 7.62)	3.75 (2.17, 6.46)	1.34 (0.69, 2.61)	5.18 (3.40, 7.89)	5.27 (3.07, 9.05)

b. Study CVIA 105 Formulation A

Serotype	GMC (95% CI), µg/mL		GMC Ratio (PCV25/PCV20) (95% CI)	GMFR (95% CI)	
	PCV25 Formulation A n=39	PCV20 n=39		PCV25 Formulation A n=39	PCV20 n=39
1	3.04 (2.30, 4.01)	6.35 (4.61, 8.76)	0.48 (0.31, 0.73)	6.39 (4.53, 9.00)	19.67 (14.81, 26.12)
3	0.76 (0.59, 0.99)	1.10 (0.82, 1.48)	0.69 (0.47, 1.02)	2.28 (1.78, 2.92)	4.23 (3.05, 5.87)
4	1.86 (1.33, 2.61)	2.66 (1.83, 3.87)	0.70 (0.43, 1.15)	12.84 (9.21, 17.92)	14.29 (9.58, 21.31)
5	5.65 (3.37, 9.45)	5.01 (3.06, 8.19)	1.13 (0.56, 2.27)	11.14 (7.64, 16.26)	11.45 (7.42, 17.66)
6B	6.08 (3.51, 10.51)	7.92 (5.11, 12.28)	0.77 (0.38, 1.53)	15.26 (9.58, 24.32)	23.01 (16.75, 31.61)
7F	3.35 (2.21, 5.06)	6.36 (4.33, 9.35)	0.53 (0.30, 0.92)	5.41 (3.93, 7.44)	10.46 (6.91, 15.84)
8	8.06 (5.77, 11.28)	7.27 (5.12, 10.33)	1.11 (0.69, 1.79)	11.00 (7.20, 16.80)	9.18 (5.79, 14.54)
9V	4.15 (2.87, 6.00)	5.41 (3.85, 7.60)	0.77 (0.47, 1.26)	11.36 (7.79, 16.58)	14.68 (10.37, 20.77)
10A	6.71 (4.31, 10.45)	12.95 (8.36, 20.06)	0.52 (0.28, 0.96)	10.58 (7.33, 15.29)	18.05 (12.68, 25.71)
12F	1.78 (1.18, 2.67)	2.11 (1.50, 2.98)	0.84 (0.50, 1.42)	9.60 (6.36, 14.48)	12.31 (8.52, 17.77)
14	9.11 (5.26, 15.78)	15.52 (9.03, 26.66)	0.59 (0.27, 1.25)	6.29 (4.26, 9.29)	12.72 (7.77, 20.82)
15B	15.26 (11.19, 20.82)	23.47 (16.41, 33.56)	0.65 (0.41, 1.04)	12.85 (8.34, 19.82)	13.95 (9.04, 21.52)
18C	7.11 (4.72, 10.70)	10.79 (7.79, 14.94)	0.66 (0.39, 1.10)	13.43 (8.74, 20.65)	15.88 (10.66, 23.65)
19A	12.57 (8.85, 17.86)	16.47 (12.12, 22.37)	0.76 (0.48, 1.21)	7.32 (4.94, 10.86)	8.69 (5.92, 12.75)
19F	11.28 (7.73, 16.47)	10.99 (7.96, 15.19)	1.03 (0.63, 1.67)	9.57 (6.60, 13.86)	8.43 (5.94, 11.96)
22F	5.88 (4.41, 7.83)	11.93 (8.39, 16.95)	0.49 (0.32, 0.77)	8.87 (6.04, 13.02)	15.16 (10.18, 22.58)
23F	4.83 (3.31, 7.03)	9.62 (6.19, 14.94)	0.50 (0.28, 0.89)	13.91 (9.21, 20.99)	19.49 (12.55, 30.26)
33F	2.31 (1.61,3.31)	10.23 (7.16, 14.61)	0.23 (0.14, 0.37)	2.27 (1.84, 2.80)	8.80 (5.71, 13.56)
2	24.75 (17.07,35.89)	3.07 (2.10, 4.47)	8.07 (4.80, 13.60)	11.42 (7.65, 17.06)	1.20 (1.04, 1.39)
6C	22.84 (13.91,37.53)	23.32 (15.23, 35.73)	0.98 (0.51, 1.86)	11.15 (7.42, 16.75)	11.24 (7.16, 17.66)
9N	5.10 (3.45,7.53)	1.23 (0.88, 1.71)	4.15 (2.51, 6.86)	13.05 (8.18, 20.82)	2.88 (2.09, 3.97)
15A	33.58 (23.05,48.92)	9.31 (6.35, 13.66)	3.61 (2.13, 6.11)	10.47 (6.86, 15.97)	3.14 (2.35, 4.18)
16F	32.26 (22.03,47.24)	8.96 (6.52, 12.32)	3.60 (2.21, 5.87)	6.21 (4.63, 8.34)	1.74 (1.42, 2.13)
24F	36.42 (20.57,64.50)	8.22 (6.12, 11.05)	4.43 (2.35, 8.34)	7.03 (4.64, 10.63)	1.43 (1.19, 1.71)
35B	22.47 (16.26,31.03)	8.76 (6.35, 12.09)	2.56 (1.64, 4.05)	2.96 (2.16, 4.05)	1.17 (0.96, 1.43)
6A	3.65 (2.25,5.92)	10.00 (6.49, 15.41)	0.36 (0.19, 0.69)	10.01 (6.71, 14.93)	24.66 (16.37, 37.15)

c. Study CVIA 105 Formulation B

Serotype	GMC (95% CI), µg/mL		GMC Ratio (PCV25/PCV20) (95% CI)	GMFR (95% CI)	
	PCV25 Formulation B n=57	PCV20 n=39		PCV25 Formulation B n=57	PCV20 n=39
1	2.56 (1.97, 3.31)	6.35 (4.61, 8.76)	0.40 (0.27, 0.60)	6.54 (5.14, 8.33)	19.67 (14.81, 26.12)
3	0.68 (0.56, 0.84)	1.10 (0.82, 1.48)	0.62 (0.44, 0.88)	2.40 (1.95, 2.96)	4.23 (3.05, 5.87)
4	1.87 (1.39, 2.50)	2.66 (1.83, 3.87)	0.70 (0.44, 1.11)	10.54 (8.11,13.70)	14.29 (9.58, 21.31)
5	6.13 (4.15, 9.06)	5.01 (3.06, 8.19)	1.23 (0.66, 2.27)	14.62 (10.46, 20.42)	11.45 (7.42, 17.66)
6B	6.42 (4.61, 8.93)	7.92 (5.11, 12.28)	0.81 (0.48, 1.38)	18.67 (13.77, 25.30)	23.01 (16.75, 31.61)
7F	4.22 (3.22, 5.53)	6.36 (4.33, 9.35)	0.66 (0.42, 1.04)	5.67 (4.38, 7.35)	10.46 (6.91, 15.84)
8	9.02 (7.24, 11.25)	7.27 (5.12, 10.33)	1.24 (0.84, 1.83)	13.41 (10.15, 17.71)	9.18 (5.79, 14.54)
9V	4.12 (3.08, 5.51)	5.41 (3.85, 7.60)	0.76 (0.49, 1.19)	9.63 (7.57, 12.25)	14.68 (10.37, 20.77)
10A	9.02 (6.35, 12.80)	12.95 (8.36, 20.06)	0.70 (0.40, 1.21)	12.94 (9.48, 17.68)	18.05 (12.68, 25.71)
12F	1.29 (0.94, 1.77)	2.11 (1.50, 2.98)	0.61 (0.38, 0.98)	8.88 (6.62, 11.91)	12.31 (8.52, 17.77)
14	8.73 (6.15, 12.39)	15.52 (9.03, 26.66)	0.56 (0.31, 1.03)	5.86 (4.36, 7.88)	12.72 (7.77, 20.82)
15B	15.76 (11.58, 21.46)	23.47 (16.41, 33.56)	0.67 (0.42, 1.07)	10.75 (7.87, 14.67)	13.95 (9.04, 21.52)
18C	6.12 (4.59, 8.16)	10.79 (7.79, 14.94)	0.57 (0.37, 0.88)	12.29 (9.05, 16.69)	15.88 (10.66, 23.65)
19A	15.81 (11.80, 21.18)	16.47 (12.12, 22.37)	0.96 (0.62, 1.48)	8.87 (6.17, 12.75)	8.69 (5.92, 12.75)
19F	15.09 (11.34, 20.09)	10.99 (7.96, 15.19)	1.37 (0.89, 2.11)	11.35 (8.08, 15.95)	8.43 (5.94, 11.96)
22F	6.04 (4.46, 8.17)	11.93 (8.39, 16.95)	0.51 (0.32, 0.80)	9.14 (6.38, 13.11)	15.16 (10.18, 22.58)
23F	5.53 (4.03, 7.57)	9.62 (6.19, 14.94)	0.57 (0.34, 0.97)	13.14 (9.14, 18.88)	19.49 (12.55, 30.26)
33F	2.51 (1.89, 3.33)	10.23 (7.16, 14.61)	0.25 (0.16, 0.38)	2.68 (2.14, 3.36)	8.80 (5.71, 13.56)
2	20.18 (15.78, 25.80)	3.07 (2.10, 4.47)	6.58 (4.31, 10.06)	9.20 (6.83, 12.41)	1.20 (1.04, 1.39)
6C	23.96 (17.32, 33.16)	23.32 (15.23, 35.73)	1.03 (0.61, 1.73)	12.45 (8.70, 17.83)	11.24 (7.16, 17.66)
9N	4.16 (3.02, 5.74)	1.23 (0.88, 1.71)	3.39 (2.12, 5.41)	10.72 (7.91, 14.51)	2.88 (2.09, 3.97)
15A	33.89 (24.81, 46.30)	9.31 (6.35, 13.66)	3.64 (2.24, 5.92)	9.65 (7.05, 13.22)	3.14 (2.35, 4.18)
16F	28.13 (20.61, 38.40)	8.96 (6.52, 12.32)	3.14 (1.99, 4.94)	5.18 (3.78, 7.09)	1.74 (1.42, 2.13)
24F	24.16 (14.87, 39.26)	8.22 (6.12, 11.05)	2.94 (1.56, 5.52)	4.80 (3.33, 6.93)	1.43 (1.19, 1.71)
35B	17.41 (13.98, 21.68)	8.76 (6.35, 12.09)	1.99 (1.37, 2.88)	1.97 (1.58, 2.46)	1.17 (0.96, 1.43)
6A	3.82 (2.75, 5.31)	10.00 (6.49, 15.41)	0.38 (0.23, 0.65)	10.32 (7.77, 13.70)	24.66 (16.37, 37.15)

d. Study CVIA 105 Formulation C

Serotype	GMC(95% CI), µg/mL		GMC Ratio (PCV25/PCV20) (95% CI)	GMFR (95% CI)	
	PCV25 Formulation C n=79	PCV20 n=39		PCV25 Formulation C n=79	PCV20 n=39
1	4.21 (3.48,5 .09)	6.35 (4.61, 8.76)	0.66 (0.47, 0.94)	9.90 (7.77,12.61)	19.67 (14.81, 26.12)
3	0.80 (0.66, 0.98)	1.10 (0.82, 1.48)	0.73 (0.52, 1.03)	3.38 (2.75, 4.16)	4.23 (3.05, 5.87)
4	1.83 (1.45, 2.31)	2.66 (1.83, 3.87)	0.69 (0.45, 1.04)	11.09 (8.78,14.01)	14.29 (9.58, 21.31)
5	5.90 (4.33, 8.05)	5.01 (3.06, 8.19)	1.18 (0.68, 2.05)	14.91 (11.51,19.32)	11.45 (7.42, 17.66)
6B	6.22 (4.60, 8.42)	7.92 (5.11, 12.28)	0.79 (0.47, 1.33)	19.83 (14.76, 26.65)	23.01 (16.75, 31.61)
7F	4.22 (3.13, 5.67)	6.36 (4.33, 9.35)	0.66 (0.40, 1.09)	7.10 (5.53, 9.11)	10.46 (6.91, 15.84)
8	12.59 (9.98, 15.87)	7.27 (5.12, 10.33)	1.73 (1.15, 2.60)	17.67 (13.45, 23.21)	9.18 (5.79, 14.54)
9V	5.77 (4.53, 7.34)	5.41 (3.85, 7.60)	1.07 (0.70, 1.61)	14.60 (11.45, 18.62)	14.68 (10.37, 20.77)
10A	9.21 (7.15, 11.87)	12.95 (8.36, 20.06)	0.71 (0.45, 1.14)	14.14 (11.31, 17.66)	18.05 (12.68, 25.71)
12F	1.65 (1.24, 2.19)	2.11 (1.50, 2.98)	0.78 (0.49, 1.24)	10.72 (8.42, 13.67)	12.31 (8.52, 17.77)
14	11.44 (8.39, 15.60)	15.52 (9.03, 26.66)	0.74 (0.41, 1.31)	8.92 (6.93, 11.47)	12.72 (7.77, 20.82)
15B	18.21 (14.04, 23.63)	23.47 (16.41, 33.56)	0.78 (0.50, 1.21)	13.26 (10.02, 17.55)	13.95 (9.04, 21.52)
18C	6.19 (4.72, 8.13)	10.79 (7.79, 14.94)	0.57 (0.37, 0.90)	12.52 (9.47, 16.55)	15.88 (10.66, 23.65)
19A	17.72 (14.01, 22.42)	16.47 (12.12, 22.37)	1.08 (0.73, 1.60)	12.20 (9.29, 16.03)	8.69 (5.92, 12.75)
19F	15.56 (12.02, 20.14)	10.99 (7.96, 15.19)	1.42 (0.92, 2.17)	16.75 (12.82, 21.88)	8.43 (5.94, 11.96)
22F	7.16 (5.64, 9.07)	11.93 (8.39, 16.95)	0.60 (0.40, 0.91)	11.36 (8.45, 15.27)	15.16 (10.18, 22.58)
23F	8.12 (5.99, 11.03)	9.62 (6.19, 14.94)	0.84 (0.50, 1.43)	21.85 (15.90, 30.04)	19.49 (12.55, 30.26)
33F	2.56 (2.04, 3.21)	10.23 (7.16, 14.61)	0.25 (0.17, 0.37)	3.31 (2.77, 3.95)	8.80 (5.71, 13.56)
2	34.74 (28.14, 42.88)	3.07 (2.10, 4.47)	11.33 (7.63, 16.82)	20.16 (15.19, 26.74)	1.20 (1.04, 1.39)
6C	34.33 (24.82, 47.48)	23.32 (15.23, 35.73)	1.47 (0.85, 2.54)	17.84 (12.58, 25.30)	11.24 (7.16, 17.66)
9N	5.98 (4.54, 7.88)	1.23 (0.88, 1.71)	4.87 (3.09, 7.65)	14.66 (11.71, 18.36)	2.88 (2.09, 3.97)
15A	37.52 (29.00, 48.54)	9.31 (6.35, 13.66)	4.03 (2.57, 6.32)	12.16 (9.22, 16.04)	3.14 (2.35, 4.18)
16F	32.04 (24.61, 41.71)	8.96 (6.52, 12.32)	3.58 (2.32, 5.52)	6.09 (4.76, 7.78)	1.74 (1.42, 2.13)
24F	31.59 (21.33, 46.79)	8.22 (6.12, 11.05)	3.84 (2.12, 6.95)	5.62 (4.24, 7.45)	1.43 (1.19, 1.71)
35B	19.75 (16.43, 23.74)	8.76 (6.35, 12.09)	2.25 (1.60, 3.17)	3.12 (2.60, 3.74)	1.17 (0.96, 1.43)
6A	5.17 (3.70, 7.21)	10.00 (6.49, 15.41)	0.52 (0.30, 0.90)	13.12 (9.47, 18.17)	24.66 (16.37, 37.15)

CI, confidence interval; GMC, geometric mean concentration; GMFR, geometric mean fold rise.Formulation A (2.2/125): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 125 μg aluminium as aluminium phosphate Formulation B (2.2/250): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate Formulation C (4.4/250): 4.4 μg for each serotype polysaccharide (except 8.8 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate

Note: blue = common to PCV25 and PCV20; green = unique to PCV25; pink = unique to PCV20.

doses are given [34]. Neonatal dendritic cells are immature and less efficient at producing cytokines needed for promoting T-helper responses, and infant B-cells have lower levels of MHC class II expression and decreased likelihood of activation and differentiation after antigen exposure [35]. Aluminium-based adjuvants have less of an impact in adults, but may be more critical in children, for whom antigen absorption into larger particles or a depot effect may be important for improved antigen presentation and initial priming [23,36]. These many immunologic differences reduce the predictive value of adult data for infants.

In addition to age, an important limitation of the study was the lack of racial diversity and the fact that results from predominantly white Canadian participants might differ from adult data in the target LMIC populations. Another limitation is the small sample size designed for an initial assessment of safety and not statistically powered for non-inferiority comparisons of immunogenicity.

Table 5

Opsonophagocytic assay geometric mean titres and geometric mean fold rises at 28 days post-dose

a. Study CVIA 096

Serotype	GMT (95% CI)		GMT Ratio (PCV25/PCV20) (95% CI)	GMFR (95% CI)	
	PCV25 Formulation A n=30	PCV20 n=29		PCV25 Formulation A n=30	PCV20 n=29
1	96.14 (51.42, 179.74)	250.62 (131.44, 477.84)	0.38 (0.16, 0.92)	9.24 (5.16, 16.54)	20.17 (11.06, 36.77)
3	52.58 (31.86, 86.77)	118.35 (72.31, 193.72)	0.44 (0.22, 0.88	2.92 (1.99, 4.29)	7.12 (4.39, 11.54)
4	2417.88 (1849.34, 3161.21)	3390.51 (2321.35, 4952.10)	0.71 (0.45, 1.12)	42.15 (16.80, 105.78)	66.56 (23.90, 185.35)
5	1587.23 (1010.03, 2494.27)	994.56 (617.84, 1600.97)	1.60 (0.84, 3.03)	141.36 (90.10, 221.79)	133.97 (87.08. 206.12)
6B	10612.69 (8101.66, 13901.98)	9868.82 (6844.11, 14230.27)	1.08 (0.69, 1.67)	62.59 (32.10, 122.08)	62.19 (26.16, 147.89)
7F	5751.07 (4678.66, 7069.30)	10918.54 (7012.29, 17000.80)	0.53 (0.33, 0.84)	10.31 (5.07, 20.98)	16.07 (7.20, 35.86)
8	2240.36 (1734.44, 2893.85)	4236.50 (3025.11, 5932.99)	0.53 (0.35 0.80)	35.93 (13.95, 92.57)	62.24 (25.39, 152.57)
9V	2850.17 (2187.79, 3713.10)	4993.46 (3718.83, 6704.98)	0.57 (0.39, 0.84)	14.04 (6.03, 32.69)	11.62 (5.84, 23.12)
10A	16621.83 (10379.48, 26618.39)	46601.79 (26038.96, 83402.97)	0.36 (0.17, 0.74)	8.65 (4.67, 16.02)	24.78 (13.17, 46.62)
12F	9894.76 (6652.40, 14717.43)	10187.80 (7277.63, 14261.67)	0.97 (0.58, 1.62)	399.57 (168.42, 947.98)	362.47 (157.02, 836.70)
14	4398.99 (3012.97, 6422.59)	8726.07 (4883.24, 15592.98)	0.50 (0.26, 0.99)	11.88 (5.61, 25.17)	39.22 (17.85, 86.19)
15B	27589.96 (18531.17, 41077.05)	28910.04 (19723.25, 42375.89)	0.95 (0.56, 1.64)	35.16 (15.50, 79.74)	33.01 (14.42, 75.60)
18C	1550.67 (1170.17, 2054.89)	2072.86 (1469.72, 2923.51)	0.75 (0.48, 1.15)	29.02 (12.97, 64.92)	42.92 (19.30, 95.45)
19A	5611.01 (3966.78, 7936.78)	8618.19 (6809.86, 10906.71)	0.65 (0.43, 0.98)	40.93 (18.35, 91.32)	40.07 (16.84, 95.33)
19F	3540.77 (2568.69, 4880.72)	2312.09 (1708.80, 3128.38)	1.53 (0.99, 2.36)	43.28 (19.34, 96.85)	29.76 (13.38, 66.21)
22F	14765.24 (9204.58, 23685.20)	23740.19 (15314.68, 36801.07)	0.62 (0.33, 1.17)	19.86 (10.51, 37.52)	39.66 (16.67, 94.37)
23F	2097.55 (1366.14, 3220.55)	3791.17 (2471.39, 5815.74)	0.55 (0.31, 1.00)	32.99 (15.31, 71.06)	64.22 (25.26, 163.28)
33F	41971.42 (24449.63, 72050.17)	87534.68 (56099.95, 136583.4)	0.48 (0.24, 0.95)	5.03 (2.85, 8.89)	14.72 (8.85, 24.49)
2	1498.41 (1107.62, 2027.09)	340.64 (189.73, 611.57)	4.40 (2.36, 8.21)	10.24 (5.25, 19.99)	0.97 (0.70, 1.36)
6C	18455.00 (13830.38, 24625.99)	13384.01 (8614.86, 20793.36)	1.38 (0.83, 2.30)	58.80 (26.52, 130.39)	26.01 (10.82, 62.49)
9N	25927.34 (18544.39, 36249.63)	6126.60 (3939.08, 9528.94)	4.23 (2.47, 7.26)	15.01 (8.58, 26.26)	2.32 (1.68, 3.18)
15A	47398.44 (37180.91, 60423.80)	30307.96 (21800.63, 42135.15)	1.56 (1.05, 2.33)	8.34 (6.27, 11.10)	5.59 (3.66, 8.52)
16F	6114.62 (5677.59, 6585.28)	3225.73 (2302.50, 4519.14)	1.90 (1.36, 2.64)	1.92 (1.29, 2.85)	1.38 (1.07, 1.78)
24F	2296.65 (1730.96, 3047.19)	84.16 (35.94, 197.08)	27.29 (11.49, 64.79)	14.29 (6.00, 34.02)	0.70 (0.49, 1.00)
35B	5163.35 (4607.99, 5785.65)	3889.05 (2898.14, 5218.76)	1.33 (0.98, 1.80)	1.41 (1.18, 1.69)	0.96 (0.81, 1.14)
6A	5967.61 (3923.83, 9075.93)	10460.65 (6575.93, 16640.27)	0.57 (0.31, 1.05)	105.09 (48.73, 226.63)	271.07 (121.09, 606.82)

b. Study CVIA 105 Formulation A

Serotype	GMT (95% CI)		GMT Ratio (PCV25/PCV20) (95% CI)	GMFR (95% CI)	
	PCV25 Formulation A n=39	PCV20 n=39		PCV25 Formulation A n=39	PCV20 n=39
1	124.89 (77.79, 200.51)	336.00 (208.94, 540.32)	0.37 (0.19, 0.72)	7.45 (4.69, 11.84)	33.88 (21.32, 53.83)
3	149.21 (107.60, 206.90)	270.61 (210.67, 347.60)	0.55 (0.37, 0.83)	3.91 (2.73, 5.59)	9.92 (6.65, 14.80)
4	3310.64 (2255.00, 4860.46)	4961.29 (3782.33, 6507.73)	0.67 (0.42, 1.06)	82.40 (37.88, 179.25)	57.73 (26.63, 125.11)
5	841.20 (462.63, 1529.56)	667.86 (392.07, 1137.67)	1.26 (0.57, 2.77)	68.52 (42.08, 111.57)	69.14 (44.65, 107.06)
6B	5170.17 (3067.04, 8715.47)	6458.35 (4512.74, 9242.77)	0.80 (0.43, 1.49)	58.92 (27.81, 124.83)	75.06 (37.92, 148.56)
7F	3285.69 (2382.65, 4530.99)	7997.72 (5772.63, 11080.48)	0.41 (0.26, 0.64)	10.08 (4.78, 21.23)	12.77 (6.14, 26.55)
8	2640.26 (1984.50, 3512.71)	2061.49 (1523.42, 2789.59)	1.28 (0.85, 1.93)	48.65 (22.92, 103.26)	29.25 (13.38, 63.91)
9V	3890.14 (2738.00, 5527.09)	5258.10 (3736.64, 7399.06)	0.74 (0.46, 1.20)	36.09 (16.31, 79.86)	31.28 (13.61, 71.88)
10A	30587.83 (21327.40, 43869.15)	44247.89 (31544.83, 62066.47)	0.69 (0.42, 1.12)	10.99 (7.67, 15.74)	13.61 (8.69, 21.33)
12F	9577.25 (6668.27, 13755.26)	13419.88 (10133.14, 17772.68)	0.71 (0.45, 1.12)	411.70 (203.05, 834.73)	495.18 (216.85, 1130.72)
14	5866.88 (3431.50, 10030.69)	13896.98 (9436.46, 20465.95)	0.42 (0.22, 0.81)	9.77 (5.27, 18.13)	32.38 (15.47, 67.76)
15B	24500.46 (16217.95, 37012.85)	38620.16 (27640.73, 53960.83)	0.63 (0.38, 1.07)	19.27 (11.94, 31.10)	25.92 (15.23, 44.13)
18C	2978.51 (1871.59, 4740.10)	3591.41 (2629.77, 4904.68)	0.83 (0.48, 1.44)	60.99 (26.90, 138.29)	62.40 (27.77, 140.21)
19A	6043.80 (4111.64, 8883.92)	6875.42 (4984.51, 9483.64)	0.88 (0.54, 1.44)	41.14 (16.47, 102.76)	29.49 (13.97, 62.26)
19F	3500.72 (2417.67, 5068.95)	2384.06 (1825.35, 3113.77)	1.47 (0.94, 2.30)	37.89 (16.87, 85.09)	23.89 (10.55, 54.07)
22F	16086.32 (11225.90, 23051.15)	22248.72 (15880.42, 31170.79)	0.72 (0.45, 1.17)	22.03 (10.49, 46.26)	17.58 (10.31, 29.96)
23F	3064.04 (1996.45, 4702.53)	5780.10 (4385.91, 7617.46)	0.53 (0.32, 0.88)	58.53 (27.23, 125.85)	91.54 (38.93, 215.22)
33F	20523.46 (14007.51, 30070.47)	51071.30 (32840.01, 79423.76)	0.40 (0.23, 0.71)	3.36 (2.39, 4.74)	7.30 (4.60, 11.58)
2	2513.19 (1834.88, 3442.27)	384.91 (247.84, 597.78)	6.53 (3.83, 11.12)	12.79 (8.12, 20.14)	1.38 (1.17, 1.63)
6C	12309.35 (8747.47, 17321.58)	12361.70 (8829.65, 17306.65)	1.00 (0.62, 1.60)	29.86 (14.10, 63.25)	27.50 (12.89, 58.66)
9N	39549.11 (30213.59, 51769.16)	5759.51 (3989.91, 8313.97)	6.87 (4.39, 10.75)	25.01 (14.23, 43.98)	3.75 (2.54, 5.54)
15A	20104.71 (13962.24, 28949.48)	9941.25 (6536.13, 15120.32)	2.02 (1.17, 3.49)	10.60 (7.19, 15.62)	4.35 (2.94, 6.43)
16F	5545.11 (4719.32, 6515.40)	3825.65 (3129.13, 4677.20)	1.45 (1.12, 1.87)	2.52 (1.95, 3.26)	1.37 (1.07, 1.75)
24F	2949.69 (2032.99, 4279.75)	464.05 (221.98, 970.09)	6.36 (2.82, 14.33)	7.98 (4.23, 15.04)	1.10 (0.97, 1.25)
35B	5573.50 (5465.97, 5683.16)	5005.11 (4551.34, 5504.12)	1.11 (1.01, 1.23)	1.27 (1.10, 1.47)	1.07 (0.98, 1.17)
6A	3444.88 (2215.31, 5356.90)	9508.91 (6689.15, 13517.32)	0.36 (0.21, 0.63)	56.27 (27.90, 113.50)	136.45 (63.96, 291.10)

c. Study CVIA 105 Formulation B

Serotype	GMT (95% CI)		GMT Ratio (PCV25/PCV20) (95% CI)	GMFR (95% CI)	
	PCV25 Formulation B n=57	PCV20 n=39		PCV25 Formulation B n=57	PCV20 n=39
1	153.89 (103.00, 229.91)	336.00 (208.94, 540.32)	0.46 (0.25, 0.85)	10.36 (7.30, 14.70)	33.88 (21.32, 53.83)
3	142.72 (107.15, 190.11)	270.61 (210.67, 347.60)	0.53 (0.35, 0.78)	4.90 (3.65, 6.59)	9.92 (6.65, 14.80)
4	2589.80 (2020.29, 3319.85)	4961.29 (3782.33, 6507.73)	0.52 (0.36, 0.76)	65.63 (36.36, 118.47)	57.73 (26.63, 125.11)
5	1052.36 (668.64, 1656.29)	667.86 (392.07, 1137.67)	1.58 (0.79, 3.15)	100.50 (65.85, 153.37)	69.14 (44.65, 107.06)
6B	6139.39 (4673.12, 8065.73)	6458.35 (4512.74, 9242.77)	0.95 (0.61, 1.47)	64.82 (36.68, 114.55)	75.06 (37.92, 148.56)
7F	3404.93 (2689.60, 4310.50)	7997.72 (5772.63, 11080.48)	0.43 (0.29, 0.63)	5.07 (3.30, 7.78)	12.77 (6.14, 26.55)
8	2535.77 (2035.52, 3158.97)	2061.49 (1523.42, 2789.59)	1.23 (0.86, 1.76)	80.72 (46.62, 139.77)	29.25 (13.38, 63.91)
9V	3274.75 (2464.88, 4350.73)	5258.10 (3736.64, 7399.06)	0.62 (0.40, 0.97)	21.93 (12.66, 38.00)	31.28 (13.61, 71.88)
10A	20768.56 (15512.08, 27806.25)	44247.89 (31544.83, 62066.47)	0.47 (0.30, 0.73)	13.25 (8.78, 19.98)	13.61 (8.69, 21.33)
12F	9963.90 (7588.44, 13082.97	13419.88 (10133.14, 17772.68)	0.74 (0.50, 1.10)	551.77 (319.59, 952.63)	495.18 (216.85, 1130.72)
14	5876.29 (3903.91, 8845.17)	13896.98 (9436.46, 20465.95)	0.42 (0.24, 0.76)	8.25 (5.11, 13.30)	32.38 (15.47, 67.76)
15B	22062.94 (15467.32, 31471.08)	38620.16 (27640.73, 53960.83)	0.57 (0.35, 0.94)	18.37 (12.10, 27.87)	25.92 (15.23, 44.13)
18C	3238.63 (2351.95, 4459.59)	3591.41 (2629.77, 4904.68)	0.90 (0.57, 1.43)	55.12 (29.38, 103.41)	62.40 (27.77, 140.21)
19A	5485.65 (4422.24, 6804.78)	6875.42 (4984.51, 9483.64)	0.80 (0.55, 1.15)	23.55 (12.58, 44.07)	29.49 (13.97, 62.26)
19F	3418.27 (2551.52, 4579.46)	2384.06 (1825.35, 3113.77)	1.43 (0.95, 2.16)	37.35 (20.56, 67.87)	23.89 (10.55, 54.07)
22F	12828.54 (9195.78, 17896.41)	22248.72 (15880.42, 31170.79)	0.58 (0.36, 0.93)	15.39 (9.33, 25.38)	17.58 (10.31, 29.96)
23F	3630.62 (2756.37, 4782.17)	5780.10 (4385.91, 7617.46)	0.63 (0.42, 0.93)	40.72 (21.74, 76.26)	91.54 (38.93, 215.22)
33F	22832.36 (16428.77, 31731.94)	51071.30 (32840.01, 79423.76)	0.45 (0.26, 0.76)	3.41 (2.51, 4.64)	7.30 (4.60, 11.58)
2	2633.60 (2191.05, 3165.55)	384.91 (247.84, 597.78)	6.84 (4.49, 10.42)	12.38 (8.39, 18.28)	1.38 (1.17, 1.63)
6C	12362.08 (9266.23, 16492.25)	12361.70 (8829.65, 17306.65)	1.00 (0.64, 1.55)	15.90 (9.39, 26.90)	27.50 (12.89, 58.66)
9N	31023.77 (23944.52, 40196.02)	5759.51 (3989.91, 8313.97)	5.39 (3.50, 8.28)	16.14 (11.96, 21.79)	3.75 (2.54, 5.54)
15A	19108.88 (14061.19, 25968.58)	9941.25 (6536.13, 15120.32)	1.92 (1.17, 3.17)	7.71 (5.69, 10.45)	4.35 (2.94, 6.43)
16F	6057.22 (5698.13, 6438.93)	3825.65 (3129.13, 4677.20)	1.58 (1.32, 1.89)	2.35 (1.69, 3.28)	1.37 (1.07, 1.75)
24F	3519.95 (2994.08, 4138.18)	464.05 (221.98, 970.09)	7.59 (4.03, 14.27)	9.60 (5.38, 17.14)	1.10 (0.97, 1.25)
35B	5559.27 (5422.20, 5699.81)	5005.11 (4551.34, 5504.12)	1.11 (1.02, 1.21)	1.29 (1.13, 1.48)	1.07 (0.98, 1.17)
6A	3976.99 (2700.43, 5857.00)	9508.91 (6689.15, 13517.32)	0.42 (0.24, 0.72)	43.33 (23.99, 78.27)	136.45 (63.96, 291.10)

d. Study CVIA 105 Formulation C

Serotype	GMT (95% CI)		GMT Ratio (PCV25/PCV20) (95% CI)	GMFR (95% CI)	
	PCV25 Formulation C n=79	PCV20 n=39		PCV25 Formulation C n=79	PCV20 n=39
1	195.99 (141.22, 272.01)	336.00 (208.94, 540.32)	0.58 (0.33, 1.03)	13.46 (9.82, 18.44)	33.88 (21.32, 53.83)
3	215.19 (164.56, 281.40)	270.61 (210.67, 347.60)	0.80 (0.52, 1.21)	7.25 (5.26, 9.99)	9.92 (6.65, 14.80)
4	3108.44 (2430.57, 3975.37)	4961.29 (3782.33, 6507.73)	0.63 (0.42, 0.93)	103.74 (63.62, 169.17)	57.73 (26.63, 125.11)
5	984.46 (703.88, 1376.89)	667.86 (392.07, 1137.67)	1.47 (0.81, 2.68)	123.80 (89.16, 171.91)	69.14 (44.65, 107.06)
6B	6149.53 (4717.58, 8016.11)	6458.35 (4512.74, 9242.77)	0.95 (0.61, 1.49)	81.09 (51.91, 126.69)	75.06 (37.92, 148.56)
7F	4353.52 (3552.47, 5335.21)	7997.72 (5772.63, 11080.48)	0.54 (0.38, 0.78)	13.36 (7.92, 22.54)	12.77 (6.14, 26.55)
8	3553.26 (2918.11, 4326.65)	2061.49 (1523.42, 2789.59)	1.72 (1.22, 2.44)	74.12 (45.78, 120.00)	29.25 (13.38, 63.91)
9V	4913.62 (3963.69, 6091.19)	5258.10 (3736.64, 7399.06)	0.93 (0.64, 1.37)	24.00 (14.73, 39.09)	31.28 (13.61, 71.88)
10A	21695.25 (16569.16, 28407.23)	44247.89 (31544.83, 62066.47)	0.49 (0.31, 0.77)	12.12 (9.18, 16.00)	13.61 (8.69, 21.33)
12F	8941.84 (6596.92, 12120.26)	13419.88 (10133.14, 17772.68)	0.67 (0.42, 1.07)	272.03 (153.71, 481.43)	495.18 (216.85, 1130.72)
14	8861.98 (6584.00, 11928.12)	13896.98 (9436.46, 20465.95)	0.64 (0.39, 1.05)	14.46 (9.34, 22.38)	32.38 (15.47, 67.76)
15B	29225.96 (22870.73, 37347.15)	38620.16 (27640.73, 53960.83)	0.76 (0.50, 1.15)	22.84 (15.64, 33.36)	25.92 (15.23, 44.13)
18C	3243.16 (2474.26, 4251.01)	3591.41 (2629.77, 4904.68)	0.90 (0.57, 1.43)	52.61 (30.93, 89.49)	62.40 (27.77, 140.21)
19A	6910.78 (5655.28, 8445.01)	6875.42 (4984.51, 9483.64)	1.01 (0.70, 1.44)	37.86 (22.68, 63.21)	29.49 (13.97, 62.26)
19F	4006.70 (3301.84, 4862.04)	2384.06 (1825.35, 3113.77)	1.68 (1.21, 2.34)	48.85 (29.94, 79.72)	23.89 (10.55, 54.07)
22F	13698.09 (10507.20, 17858.01)	22248.72 (15880.42, 31170.79)	0.62 (0.40, 0.96)	20.72 (12.94, 33.17)	17.58 (10.31, 29.96)
23F	4054.29 (3236.75, 5078.32)	5780.10 (4385.91, 7617.46)	0.70 (0.48, 1.02)	79.08 (47.89, 130.59)	91.54 (38.93, 215.22)
33F	26679.55 (20326.28, 35018.64)	51071.30 (32840.01, 79423.76)	0.52 (0.32, 0.85)	3.87 (3.00, 5.00)	7.30 (4.60, 11.58)
2	4035.78 (3463.44, 4702.72)	384.91 (247.84, 597.78)	10.48 (7.23, 15.21)	22.63 (15.90, 32.21)	1.38 (1.17, 1.63)
6C	14273.06 (11792.28, 17275.71)	12361.70 (8829.65, 17306.65)	1.15 (0.81, 1.65)	28.31 (16.92, 47.37)	27.50 (12.89, 58.66)
9N	38859.32 (31652.83, 47706.53)	5759.51 (3989.91, 8313.97)	6.75 (4.59, 9.91)	24.49 (18.06, 33.22)	3.75 (2.54, 5.54)
15A	23975.88 (18758.08, 30645.07)	9941.25 (6536.13, 15120.32)	2.41 (1.54, 3.79)	10.11 (7.87, 12.98)	4.35 (2.94, 6.43)
16F	6212.33 (5882.67, 6560.47)	3825.65 (3129.13, 4677.20)	1.62 (1.39, 1.90)	2.03 (1.76, 2.35)	1.37 (1.07, 1.75)
24F	3240.19 (2765.08, 3796.93)	464.05 (221.98, 970.09)	6.98 (4.01, 12.16)	9.68 (6.36, 14.72)	1.10 (0.97, 1.25)
35B	5565.54 (5454.49, 5678.85)	5005.11 (4551.34, 5504.12)	1.11 (1.04, 1.19)	1.39 (1.23, 1.57)	1.07 (0.98, 1.17)
6A	5787.36 (4402.56, 7607.73)	9508.91 (6689.15, 13517.32)	0.61 (0.39, 0.96)	88.41 (51.06, 153.07)	136.45 (63.96, 291.10)

CI, confidence interval; GMT, geometric mean titer; GMFR, geometric mean fold rise. Formulation A (2.2/125): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 125 μg aluminium as aluminium phosphate Formulation B (2.2/250): 2.2 μg for each serotype polysaccharide (except 4.4 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate Formulation C (4.4/250): 4.4 μg for each serotype polysaccharide (except 8.8 μg for serotype 6B) with 250 μg aluminium as aluminium phosphate

Note: blue = common to PCV25 and PCV20; green = unique to PCV25; pink = unique to PCV20.

Strengths of the studies include the excellent retention and completeness of the data and the use of established validated immunogenicity assays at the WHO Pneumococcal serology reference laboratory (IgG) enabling bridge to efficacy studies performed with earlier PCVs.

In summary, IVT PCV-25 had an acceptable safety profile and was immunogenic in healthy adults. This vaccine was designed specifically to address pneumococcal disease burden in LMICs and provides coverage against serotypes causing over 80% of the current infant disease burden across all global regions, more than twice that of PCV10 and PCV13, the currenty used vaccines in the target regions, and similar or higher than PCVs currently marketed in wealthier countries [20]. Although additional product optimization may ultimately be required to improve immunogenicity for some serotypes, adult immunogenicity data are not necessarily predictive of immunogenicity for infants.

Further development will include assessment of safety and immunogenicity of all current formulations in the target infant population with a goal to identify a well-tolerated formulation likely to meet non-inferiority criteria for currently licensed strains in addition to providing expanded coverage against novel serotypes.

CRediT authorship contribution statement

Joanne M. Langley: Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Manish Sadarangani:** Writing – review & editing, Supervision, Methodology, Investigation. **Christian Ockenhouse:** Writing – review & editing, Project administration, Methodology, Formal analysis, Conceptualization. **Luis Barreto:** Writing – review & editing, Methodology, Conceptualization. **Lingyun Ye:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Yuxiao Tang:** Writing – review & editing, Formal analysis, Conceptualization. **Janis L. Breeze:** Writing – review & editing, Supervision, Formal analysis. **Jodi Feser:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Data curation, Conceptualization. **Nancy A. Hosken:** Writing – review & editing, Methodology, Conceptualization. **Indah Andi-Lolo:** Writing – review & editing, Project administration. **Sybil A. Tasker:** Writing – review & editing, Project administration, Conceptualization. **Scott A. Halperin:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Joanne M. Langley, Manish Sadarangani, Christian Ockenhouse, Luis Barreto, Lingyun Ye, Yuxiao Tang, Janis L. Breeze, Jodi Feser, Nancy A. Hosken, Indah Andi-Lolo, Scott A. Halperin reports financial support, article publishing charges, and writing assistance were provided by Inventprise. Sybil Tasker reports article publishing charges, travel, and writing assistance were provided by Inventprise. Scott Halperin, reports a relationship with Merck & Co Inc. that includes: funding grants. Scott Halperin reports a relationship with Pfizer that includes: funding grants. Scott Halperin reports a relationship with GSK Vaccines that includes: consulting or advisory. Scott Halperin reports a relationship with Moderna Inc. that includes: consulting or advisory. Scott Halperin reports a relationship with AstraZeneca that includes: consulting or advisory. Scott Halperin reports a relationship with Seqirus Inc. that includes: consulting or advisory. Scott Halperin reports a relationship with Aramis that includes: consulting or advisory. Luis Barreto reports a relationship with CanSino Biologics Inc. that includes: consulting or advisory. Luis Barreto reports a relationship with Medicago Inc. that includes: consulting or advisory. Joanne M. Langley reports a relationship with Merck & Co Inc. that includes: funding grants. Joanne M Langley reports a relationship with GSK Vaccines that includes: consulting or advisory. Joanne M Langley reports a relationship with Vaxcyte Inc. that includes: consulting or advisory. Joanne M Langley reports a relationship with Meningitis Foundation of Canada that includes: consulting or advisory. Manish Sadarangani reports a relationship with Merck & Co Inc. that includes: funding grants. Manish Sadarangani reports a relationship with Moderna Inc. that includes: funding grants. Manish Sadarangani reports a relationship with Pfizer that includes: funding grants. Manish Sadarangani reports a relationship with GlaxoSmithKline that includes: funding grants. Manish Sadarangani reports a relationship with Sanofi Pasteur Inc. that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.vaccine.2026.128236.

Data availability

Inventprise is committed to providing access to anonymised data from clinical trials for legitimate scientific research. Requests should be addressed to info@inventprise.com and must be accompanied by a detailed analysis plan, which will be reviewed for scientific validity. Sharing of data may require execution of a data sharing agreement.

References

[1] American Academy of Pediatrics. ACIP, Academy recommendations for PCV7 almost identical. https://publications.aap.org/aapnews/article-abstract/17/2/52/18636/ACIP-Academy-recommendations-for-PCV7-almost; 2025. 01 August 2000. [Accessed 25 June 2025].

[2] View-Hub by IVAC. Pneumococcal conjugate vaccine (PCV): Introduction & use. https://view-hub.org/vaccine/pcv?set=current-dosing-schedule&group=vaccine-introduction&category=pcv; 2025 [Accessed 27 May 2025].

[3] Rodgers GL, Whitney CG, Klugman KP. Triumph of pneumococcal conjugate vaccines: overcoming a common foe. J Infect Dis 2021;224:S352–9. https://doi.org/10.1093/infdis/jiaa535.

[4] Reyburn R, Tsatsaronis A, von Mollendorf C, Mulholland K, Russell FM. ARI Review Group. Systematic review on the impact of the pneumococcal conjugate vaccine ten valent (PCV10) or thirteen valent (PCV13) on all-cause, radiologically confirmed and severe pneumonia hospitalisation rates and pneumonia mortality in children 0-9 years old. J Glob Health 2023;13:05002. https://doi.org/10.7189/jogh.13.05002.

[5] Berman-Rosa M, O'Donnell S, Barker M, Quach C. Efficacy and effectiveness of the PCV-10 and PCV-13 vaccines against invasive pneumococcal disease. Pediatrics 2020;145(4). https://doi.org/10.1542/peds.2019-0377. e20190377. Apr.

[6] Loughran AJ, Orihuela CJ, Tuomanen EI. Streptococcus pneumoniae: invasion and inflammation. Microbiol Spectr 2019;7:10. https://doi.org/10.1128/microbiolspec.GPP3-0004-2018.

[7] GBD. Antimicrobial Resistance Collaborators. Global mortality associated with 33 bacterial pathogens in 2019: A systematic analysis for the global burden of disease study 2019. Lancet 2019;2022(400):2221–48. https://doi.org/10.1016/S0140-6736(22)02185-7.

[8] Løchen A, Croucher NJ, Anderson RM. Divergent serotype replacement trends and increasing diversity in pneumococcal disease in high income settings reduce the benefit of expanding vaccine valency. Sci Rep 2020;10:18977. https://doi.org/10.1038/s41598-020-75691-5.

[9] Du QQ, Shi W, Yu D, Yao KH. Epidemiology of non-vaccine serotypes of Streptococcus pneumoniae before and after universal administration of pneumococcal conjugate vaccines. Hum Vaccin Immunother 2021;17:5628–37. https://doi.org/10.1080/21645515.2021.1985353.

[10] Saha SK, Al Emran HM, Hossain B, Darmstadt GL, Saha S, Islam M, et al. Streptococcus pneumoniae serotype-2 childhood meningitis in Bangladesh: a newly recognized pneumococcal infection threat. PloS One 2012;7:e32134. https://doi.org/10.1371/journal.pone.0032134.

[11] Gaensbauer JT, Asturias EJ, Soto M, Holt E, Olson D, Halsey NA, et al. Pediatric invasive pneumococcal disease in Guatemala City: importance of serotype 2. Pediatr Infect Dis J 2016;35:e139–43. https://doi.org/10.1097/INF.0000000000001067.

[12] Nunes S, Valente C, Sá-Leão R, Lencastre H. Temporal trends and molecular epidemiology of recently described serotype 6C of Streptococcus pneumoniae. J Clin Microbiol 2009;47(2):472–4. https://doi.org/10.1128/JCM.01984-08.
[13] Amin-Chowdhury Z, Collins S, Sheppard C, Litt D, Fry NK, Andrews N, et al. Characteristics of invasive pneumococcal disease caused by emerging serotypes after the introduction of the 13-valent pneumococcal conjugate vaccine in England: a prospective observational cohort study, 2014-2018. Clin Infect Dis 2020;71: e235–43. https://doi.org/10.1093/cid/ciaa043.
[14] Sheppard C, Fry NK, Mushtaq S, Woodford N, Reynolds R, Janes R, et al. Rise of multidrug-resistant non-vaccine serotype 15A Streptococcus pneumoniae in the United Kingdom, 2001 to 2014. Euro Surveill 2016;21:30423. https://doi.org/10.2807/1560-7917.ES.2016.21.50.30423.
[15] Nakano S, Fujisawa T, Ito Y, Chang B, Matsumura Y, Yamamoto M, et al. Spread of meropenem-resistant Streptococcus pneumoniae serotype 15A-ST63 clone in Japan, 2012-2014. Emerg Infect Dis 2018;24:275–83. https://doi.org/10.3201/eid2402.171268.
[16] Marsh RL, Smith-Vaughan H, Beissbarth J, Hare K, Kennedy M, Wigge C, et al. Molecular characterisation of pneumococcal serotype 16F: established predominant carriage and otitis media serotype in the 7vPCV era. Vaccine 2007; 25:2434–6. https://doi.org/10.1016/j.vaccine.2006.09.016.
[17] Ouldali N, Levy C, Varon E, Bonacorsi S, Béchet S, Cohen R, et al. Incidence of paediatric pneumococcal meningitis and emergence of new serotypes: a time-series analysis of a 16-year French national survey. Lancet Infect Dis 2018;18:983–91. https://doi.org/10.1016/S1473-3099(18)30349-9.
[18] Lo SW, Mellor K, Cohen R, Alonso AR, Belman S, Kumar N, et al. Emergence of a multidrug-resistant and virulent Streptococcus pneumoniae lineage mediates serotype replacement after PCV13: an international whole-genome sequencing study. Lancet Microbe 2022;3:e735–43. https://doi.org/10.1016/S2666-5247(22)00158-6.
[19] Olarte L, Kaplan SL, Barson WJ, Romero JR, Lin PL, Tan TQ, et al. Invasive serotype 35B pneumococci including an expanding serotype switch lineage. Emerging Infect Dis 2018;24:405. https://doi.org/10.3201/eid2402.170982.
[20] Quesada MG, Peterson ME, Bennett JC, et al. Serotype distribution of remaining invasive pneumococcal disease after extensive use of ten-valent and 13-valent pneumococcal conjugate vaccines (the PSERENADE project): a global surveillance analysis. Lancet Infect Dis 2025;25:445–6. https://doi.org/10.1016/S1473-3099(24)00588-7.
[21] Yildirim I, Hanage WP, Lipsitch M, Shea KM, Stevenson A, Finkelstein J, et al. Serotype specific invasive capacity and persistent reduction in invasive pneumococcal disease. Vaccine 2010;29:283–8. https://doi.org/10.1016/j.vaccine.2010.10.032.
[22] Datta A, Kapre K, Andi-Lolo I, Kapre S. Multi-valent pneumococcal conjugate vaccine for global health: from problem to platform to production. Hum Vaccin Immunother 2022;18:2117949. https://doi.org/10.1080/21645515.2022.2117949.
[23] HogenEsch H, O'Hagan DT, Fox CB. Optimizing the utilization of aluminum adjuvants in vaccines: you might just get what you want. NPJ Vaccines 2018;3:51. https://doi.org/10.1038/s41541-018-0089-x.
[24] Food and Drug Administration. Guidance for industry: Toxicity grading scale for healthy adult and adolescent volunteers enrolled in preventive vaccine clinical trials. Rockville, MD: U.S. Department of Health and Human Services; 2007. Available at: https://www.fda.gov/media/73679/download.
[25] Wernette CM, Frasch CE, Madore D, et al. Enzyme-linked immunosorbent assay for quantitation of human antibodies to pneumococcal polysaccharides. Clin Diagn Lab Immunol 2003;10(4):514–9. https://doi.org/10.1128/cdli.10.4.514-519.2003.
[26] Marchese RD, Puchalski D, Miller P, et al. Optimization and validation of a multiplex, electrochemiluminescence-based detection assay for the quantitation of immunoglobulin G serotype-specific antipneumococcal antibodies in human serum. Clin Vaccine Immunol 2009;16:387–96. https://doi.org/10.1128/CVI.00415-08.
[27] Nolan KM, Zhang Y, Antonello JM, et al. Enhanced antipneumococcal antibody electrochemiluminescence assay: validation and bridging to the WHO reference ELISA. Bioanalysis 2020;12:1363–75. https://doi.org/10.4155/bio-2020-0023.
[28] World Health Organization. Recommendations to assure the quality, safety and efficacy of pneumococcal conjugate vaccines, Annex 3, TRS No 977. Replacement of WHO TRS No 927, Annex 2. 19 October 2013. Available at: https://www.who.int/publications/m/item/pneumococcal-conjugate-vaccines-annex3-trs-977; 2025.
[29] Burton RL, Nahm MH. Development of a fourfold multiplexed opsonophagocytosis assay for pneumococcal antibodies against additional serotypes and discovery of serological subtypes in Streptococcus pneumoniae serotype 20. Clin Vaccine Immunol 2012;19:835–41. https://doi.org/10.1128/CVI.00086-12.
[30] Senders S, Klein NP, Tamimi N, et al. A phase three study of the safety and immunogenicity of a four-dose series of 20-valent pneumococcal conjugate vaccine in healthy infants. Pediatr Inf Dis J 2024;43(6):596–603. https://doi.org/10.1097/INF.0000000000004334.
[31] Pichichero ME. Protein carriers of conjugate vaccines: characteristics, development, and clinical trials. Hum Vaccin Immunother 2013;9:2505–23. https://doi:https://doi.org/10.4161/hv.26109.
[32] Tricarico S, McNeil HC, Cleary DW, et al. Pneumococcal conjugate vaccine implementation in middle-income countries. Pneumonia (Nathan) 2017;9:6. https://doi.org/10.1186/s41479-017-0030-5.
[33] Pieren DKJ, Boer MC, de Wit J. The adaptive immune system in early life: the shift makes it count. Front Immunol 2022;13:1031924. https://doi.org/10.3389/fimmu.2022.1031924.
[34] Desikan R, Germani M, van der Graaf PH, Magee M. A quantitative clinical pharmacology-based framework for model-informed vaccine development. J Pharm Sci 2024;113:22–32. https://doi.org/10.1016/j.xphs.2023.10.043.
[35] Pichichero ME. Challenges in vaccination of neonates, infants and young children. Vaccine 2014;32:3886–94. https://doi.org/10.1016/j.vaccine.2014.05.00.
[36] Nanishi E, Angelidou A, Rotman C, Dowling DJ, Levy O, Ozonoff A. Precision vaccine adjuvants for older adults: a scoping review. Clin Infect Dis 2022;75(Suppl. 1):S72–80. https://doi.org/10.1093/cid/ciac302.